

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT2 — VISUALIZATION DEVELOPMENT EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Development (BL4)	Assessment:	Assessment Tool 2 – Visualization Development
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO2 Statement:	<i>Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight.(BL4)</i>		
Student Name:	Tejaswini P	Reg. No.:	192324046
Section / Batch:	2023/B tech AIDS	Date:	30/7/2026

Code of Conduct

I Tejaswini P/192324046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tejaswini

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

Annexure A – VISUALIZATION DEVELOPMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Visualization Development Exercise

CO2 · BT4: Analyze ·

This rubric is used to assess student responses to the CO2-AT2 Visualization Development Exercise, in which students build a small multi-chart dashboard from given data, add at least one annotation, and write a narrative takeaway.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of which chart types to combine, how they relate to each other, and how annotation and narrative should be used to communicate insight.	Good understanding of chart combination and annotation/narrative technique, with only minor gaps.	Basic understanding of dashboard development, but with some misconceptions about which charts to combine or how to annotate them.	Poor understanding of core development concepts; combines charts or annotations that do not suit the data or the story being told.
Explanation	25%	The annotation and narrative takeaway are precise, well-justified, and clearly tied to a specific, correctly identified data point or pattern.	The annotation and narrative are clear and reasonably well-justified, with adequate connection to the data.	The annotation or narrative is present but vague, generic, or weakly connected to the actual data shown.	Annotation or narrative is missing, incorrect, or unconnected to the data in the dashboard.
Application	20%	Effectively applies multi-chart coordination, annotation, and storytelling techniques to produce a complete, cohesive dashboard for the scenario.	Applies development techniques to produce a workable dashboard, with only minor errors or missed refinements.	Limited application; the dashboard partially addresses the scenario but charts feel disconnected or the story is unclear.	Fails to apply appropriate development techniques; the charts, annotation, and narrative do not work together as a dashboard.
Organization	15%	Charts are clearly coordinated	Charts are organized and mostly coordinated,	Basic structure; charts are present but poorly aligned	Poorly organized; charts are not

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		(aligned axes or shared theme), laid out logically, with the annotation and narrative placed for maximum clarity.	with only minor issues in alignment or layout.	or the annotation/narrative feels disconnected from the layout.	coordinated, and the annotation or narrative is missing or misplaced.
Accuracy	10%	All chart values, scales, and the annotated data point are completely accurate and correctly reflect the given data.	Mostly accurate, with only minor omissions or a small error in one chart or the annotation.	Partially correct; some plotted values or the annotated point are missing or incorrect.	Largely inaccurate; major errors in plotted values, scales, or the annotated data point.

Scoring Guide

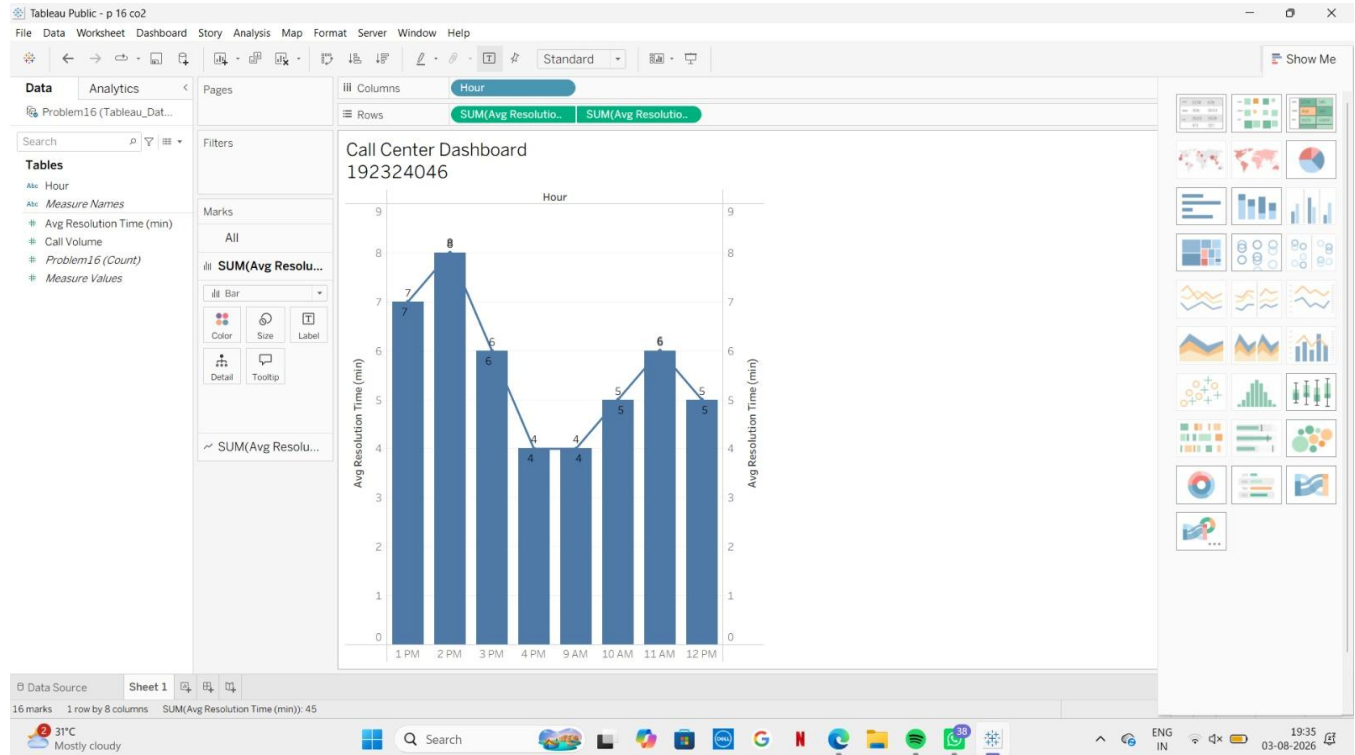
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 16 — Call Center Dashboard

Data: Hourly call volume and average resolution time (8-hour day): 9am=40/4min, 10am=55/5, 11am=70/6, 12pm=50/5, 1pm=65/7, 2pm=80/8, 3pm=60/6, 4pm=45/4.

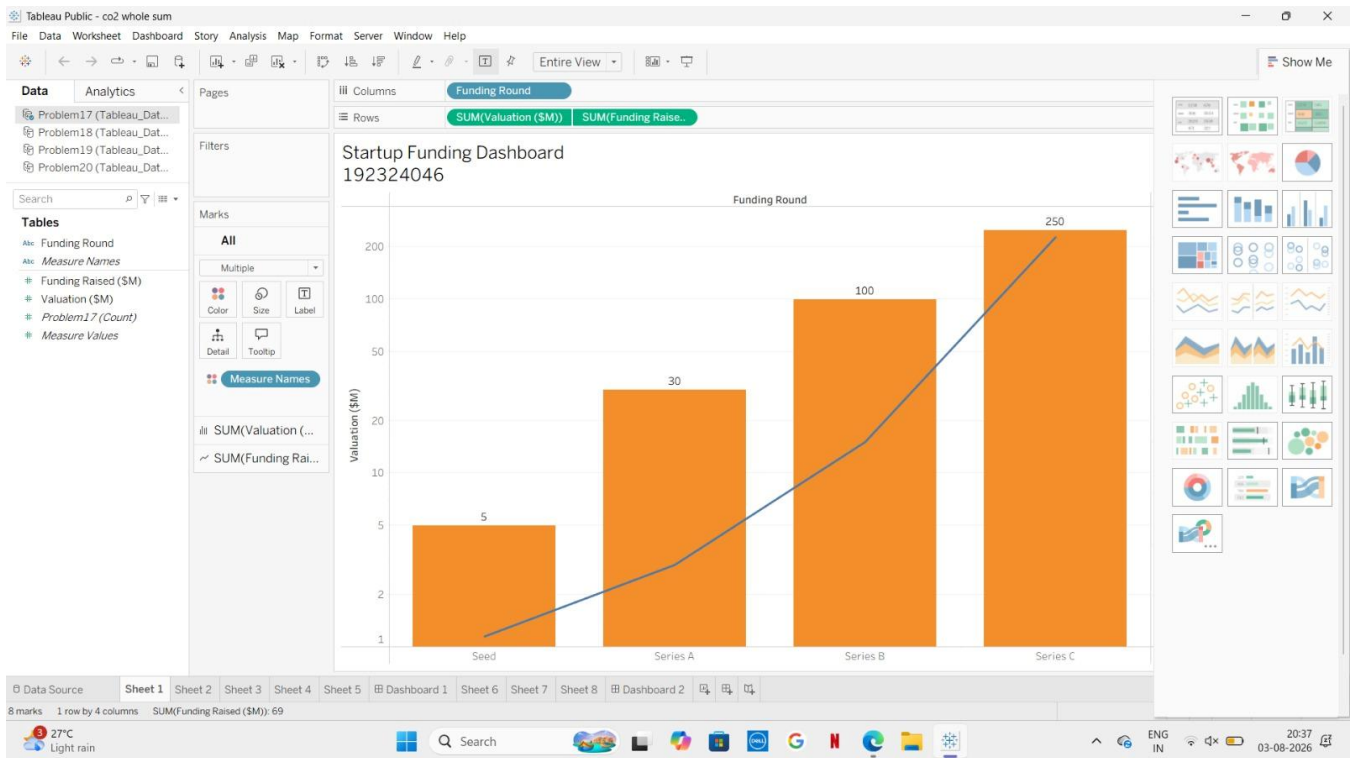
Task: Build a dashboard with a bar chart of call volume and a line chart of resolution time across the same hours. Annotate the peak-stress hour needing more staff, and write a takeaway for the call center manager.



Problem 17 — Startup Funding Dashboard

Data: Funding raised and valuation by round: Seed = \$1M / \$5M valuation, Series A = \$8M / \$30M, Series B = \$20M / \$100M, Series C = \$40M / \$250M.

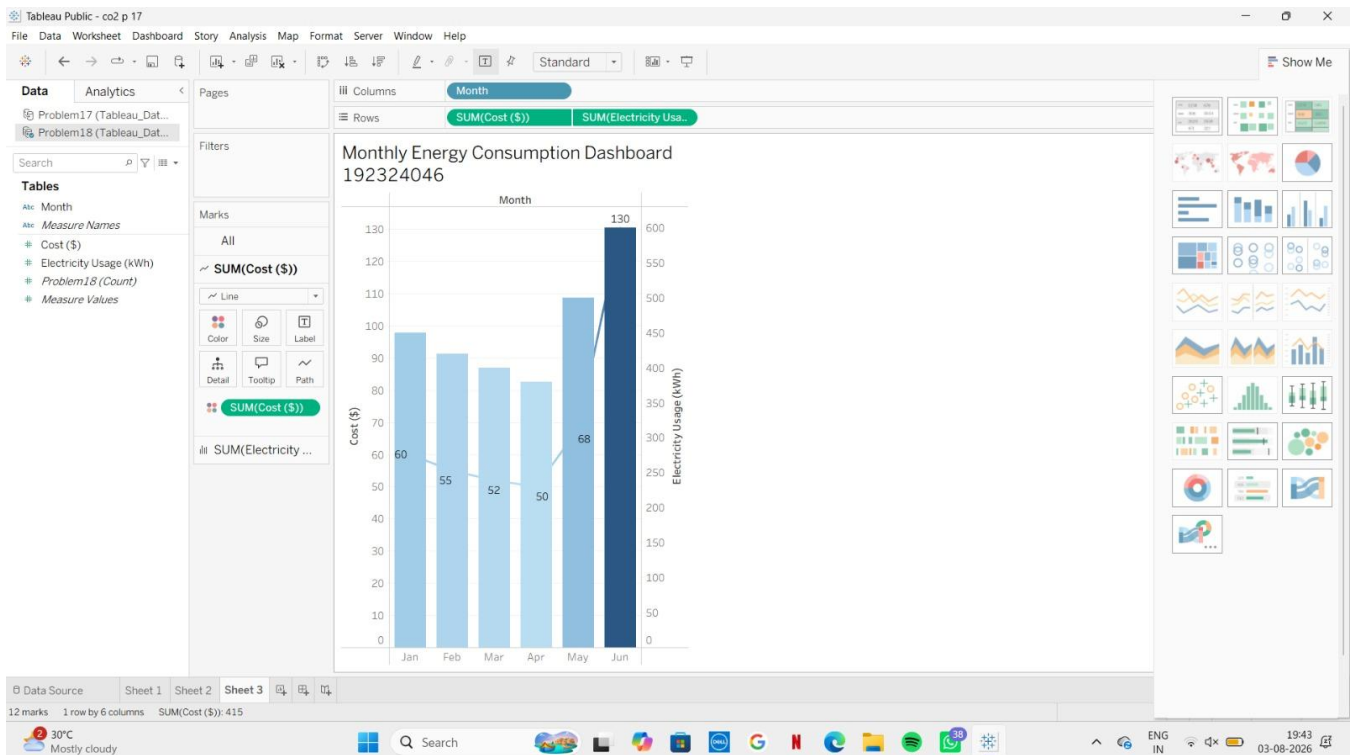
Task: Build a dashboard with a bar chart of funding raised and a line chart of valuation across the four rounds, using a logarithmic scale for valuation. Annotate the round with the sharpest valuation jump, and write a narrative for prospective investors.



Problem 18 — Energy Consumption Dashboard

Data: Monthly electricity usage (kWh) and cost (\$): Jan=450/60, Feb=420/55, Mar=400/52, Apr=380/50, May=500/68, Jun=600/130.

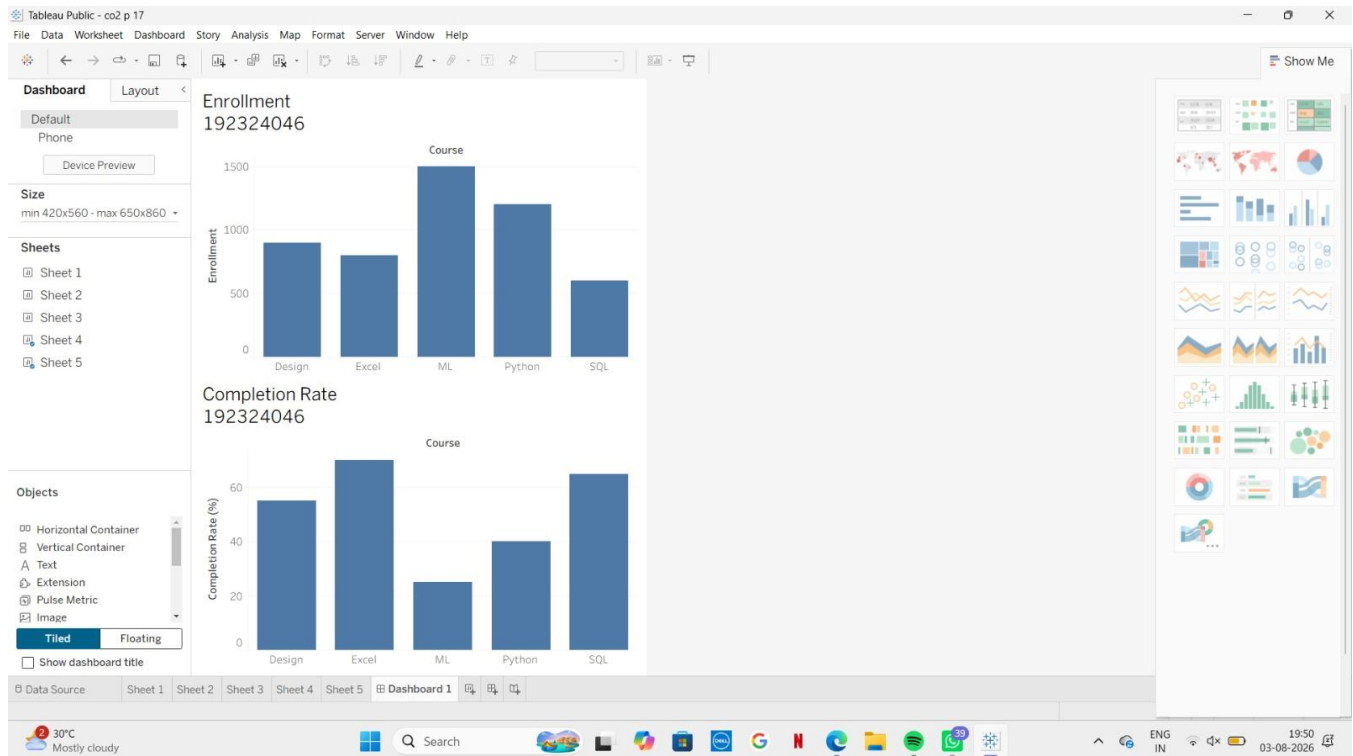
Task: Build a dashboard with a bar chart of usage and a line chart of cost for the same months. Annotate the month with an unusual cost spike relative to usage, and write a takeaway suggesting a possible cause (e.g., a rate change).



Problem 19 — Online Course Dashboard

Data: Enrollment and completion rate for 5 courses: Python = 1,200 / 40%, Excel = 800 / 70%, ML = 1,500 / 25%, SQL = 600 / 65%, Design = 900 / 55%.

Task: Build a dashboard with a bar chart of enrollment and a bar chart of completion rate for the same courses. Annotate the course with high enrollment but poor completion, and write a narrative recommending a course-redesign priority.



Problem 20 — Airline On-Time Performance Dashboard

Data: On-time percentage and passenger volume by route: NYC–LA = 75% / 5,000 pax, NYC–Chicago = 90% / 3,000, NYC–Miami = 60% / 4,500, NYC–Boston = 95% / 1,500.

Task: Build a dashboard with a bar chart of passenger volume and a bar chart of on-time percentage for the same routes. Annotate the route with the worst combination of high volume and poor punctuality, and write a takeaway for operations.

